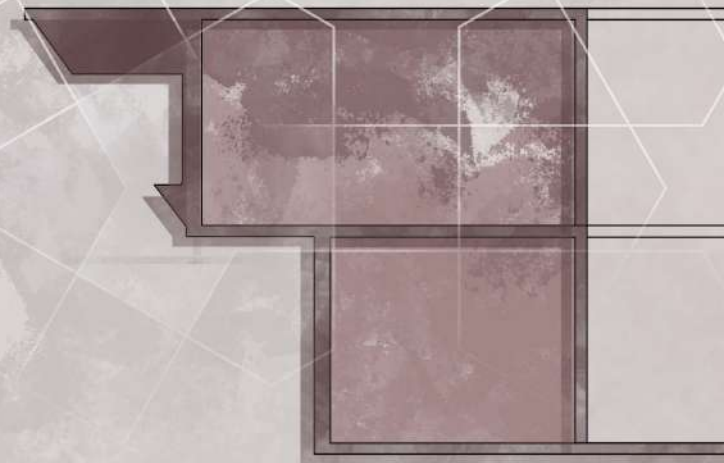


KHALED RAGAB



PORTFOLIO



KHALED RAGAB

khaled.ragab.khalil96@gmail.com
+201091158881
<https://www.linkedin.com/in/khaled-ragab/>

EDUCATION:

2014-2019 | Bachelor's Degree in Architectural Engineering
Assiut University

Designed a Cultural Centre in Baris Oasis
(Hassan Fathy's Village) as a Graduation Project.

Mapped Kharga Oasis using GIS as part
of the Graduation Project.

Professional Experience:

2021 | **Interdisciplinary Architect**
CLUSTER
(Cairo Lab for Urban Studies, Training and Environmental Research)

Coordinating between the design office and the fabrication
lab (Al-Fabrika) as a link between design and execution.

Designing and implementing 2D/3D designs and models.

Working closely with the local community and partners
in "Ard El-Liwa" (informal neighborhood in Cairo)

Manging internal information system.

Manging internal information system.

Engaging in design projects that address
urban challenges, such as housing, transportation
and community spaces.

raise awareness about urban issues through exhibitions
workshops, and public events.

March-Nov
2023 | **Archiving Specialist**
BIAS-AME
Built-environment Institute for Applied Studies - Africa and Middle East

Organizing and maintaining records systematically.

Developing classification systems to facilitate
easy access to archival materials.

2022 | **Freelance GIS Specialist**

Georeferencing and organizing different maps.

Digitizing historical maps and creating new maps

Internship:

JUN-NOV
2021 | **Interdisciplinary Architect**
Al-Qalam
(Cairo Lab for Urban Studies, Training and Environmental Research)

Engaging with local communities to learn about their
perceptions of calligraphy, its role in their lives
and its impact on urban aesthetics.

Enriching tours by Exploring how calligraphy intersects
with architecture, public signage, or street art.

Facilitating collaborations between Al-Qalam and other
urban-focused institutions, fostering interdisciplinary projects

2023 | **Outreach Team Member**
SASA
Save Ancient Studies Alliance

Communicating with potential Arabic-speaking partners

Helping with translating the website

Volunteering:

2023 | SASA (Save Ancient Studies Alliance)
Outreach Team Member, Translator

2017 | AZAD no MANGA
Editor and Art Director

Skills:

AutoCAD 2D

Adobe Suite
(Photoshop, Illustrator, InDesign)

ArcGIS

Lumion

SketchUp

Blender

Audacity

UltiMaker Cura

CONTENT	01	02	03	04	05	06
	GREEN ATLAS		FABRICATIONS		SOUND MAPPING	
		HASSAN FATHY CULTURAL CENTRE		ASYUT URBAN HERITAGE		PERSONAL PROJECTS

01

GREEN ATLAS





CONTEXT

In the light of recent developments, where the historic Orman Garden, a 150-year-old green oasis in Egypt, faces closure alongside the zoo, the Mapping Trees Initiative orchestrated a workshop. Guided by Prof. Nabeel Elhady and Prof. Rim Samir Hamdy from Cairo University, our mission was clear: map and document the trees within the vicinity of the Faculty of Engineering, Cairo University, Giza Zoo, and the cherished Orman Garden.

As a member of the team assigned to map the area around the Faculty of Engineering and Cairo University, I embarked on fieldwork that led us through the enchanting paths of the Orman Garden and the vibrant spaces of the Giza Zoo.



IDEATION

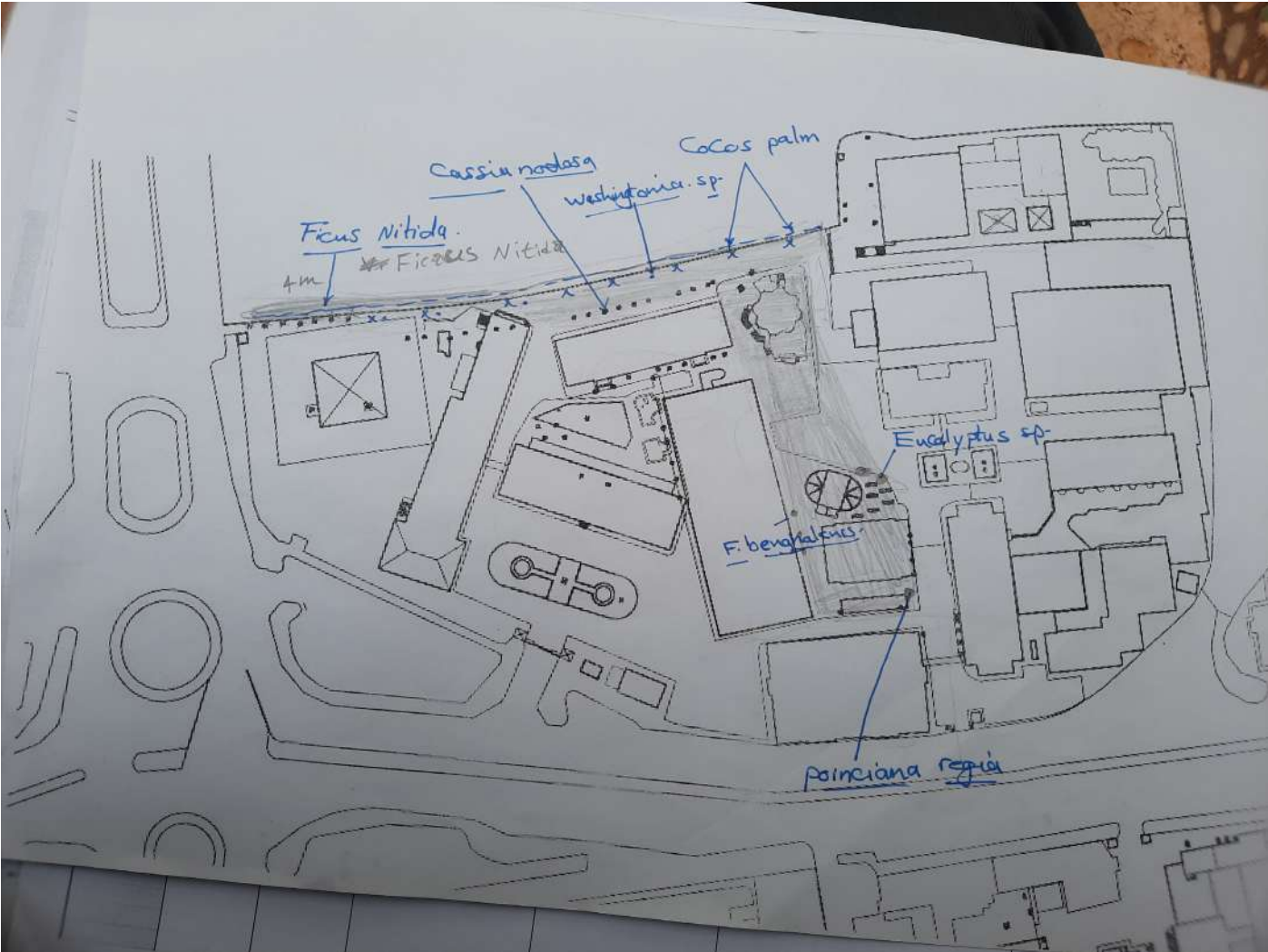
My personal motivation for participating in the workshop was to gain a better understanding of the role and agency of trees in our urban environment. I wanted to view the city from a different perspective - a “non-human” one and explore how it shapes my understanding to the urban landscape.

I started with Mr. Magdy in a team and planned the methodology that we were going to use.

We identified our zones and the paths that we would go through, prepared the sheets in which we would fill the data, and tested the application that we would use to add the longitude and latitude coordinates of each tree.

Code = E

No.	Canopy	ID	Picture of tree (with human scale)		Details Picture			Location	
			Height	Picture	Seed	Leaf	Flower	x	y
1	3m	LOCUS palmosa	6m					31.212475	30.02
2	3m	" "	6m						
3	3m	" "	6m						
4	2m	" "	4m						
5	2m	" "	4m						
6	4m	" "	9m						
7	3m	" "	4m						
8	4m	" "	4m						
9	5m	" "	5m						
10	4m	" "	5m						
11	3m	" "	2.5m						
12	3m	LOCUS palmosa	4m						
13	4m	" "	5m						
14	3m	" "	6m						
15	4m	" "	4m						
16	5m	" "	5m						
17	4m	" "	4m						
18	5m	" "	5m						
19	5m	" "	5m						
20	5m	" "	5m						
21	5m	" "	14m 1.5m						
22	4m	" "	6m						
23	4m	" "	7m						
24	4m	" "	6m						
25	4m	" "	7m						
26	5m	" "	8m						
27	5m	" "	4.5m						

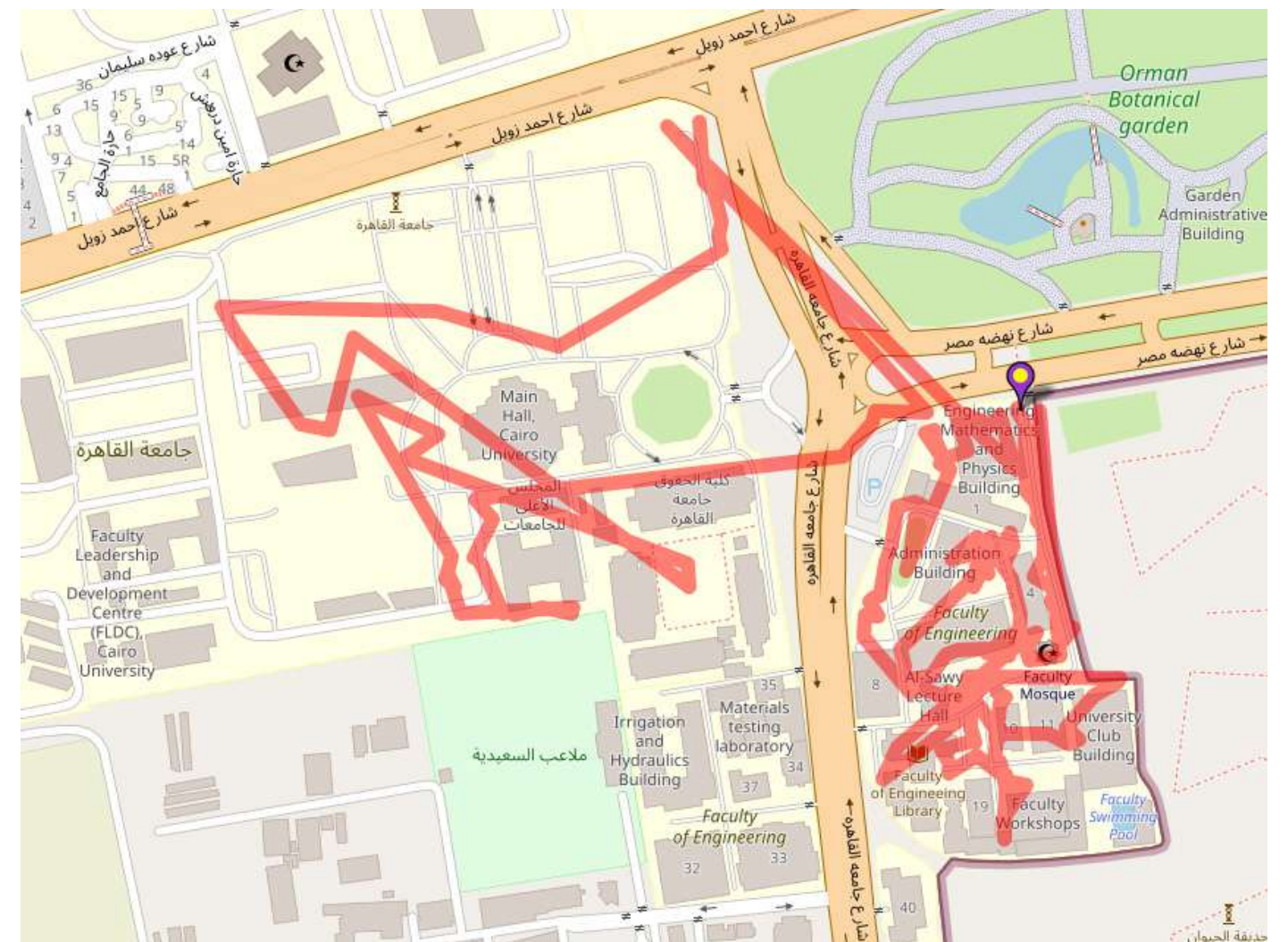
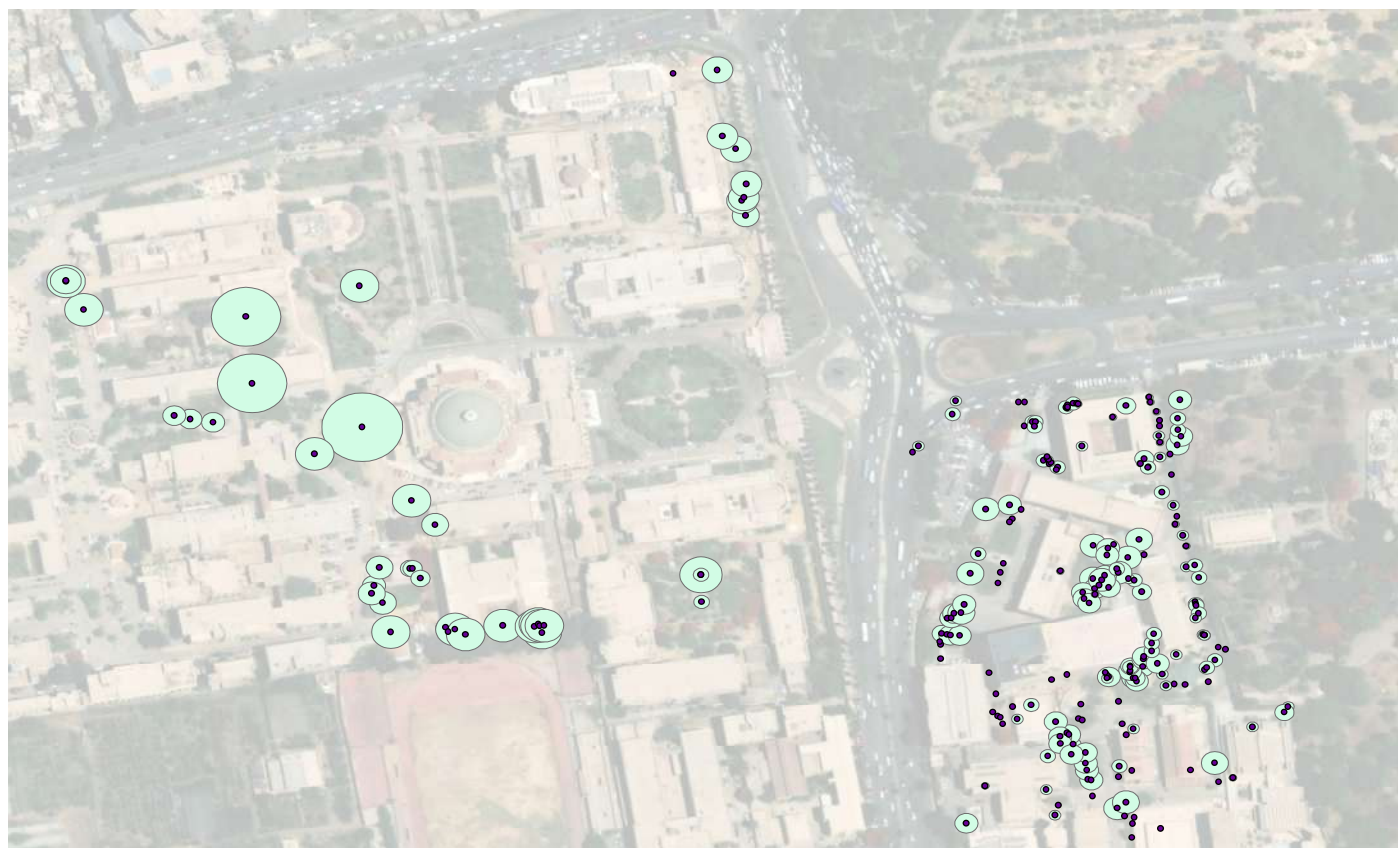




IMPLEMENTATION

After gathering the data and delivering it, I became interested in advancing the work further. I began digitizing the data.

The digitizing process involved linking images taken with their geolocation, adding attribute data for each tree, and visualizing the outcomes.



IMPLEMENTATION

After digitizing the data i had the oppertunity to participate in “forcities” exhibition in vitnam under the theme “more than human scale”.

facing the challenge of making the map accessable i used “my google maps” to make it accessable and online so anyone can add upon it and i called it “Green Atlas”.

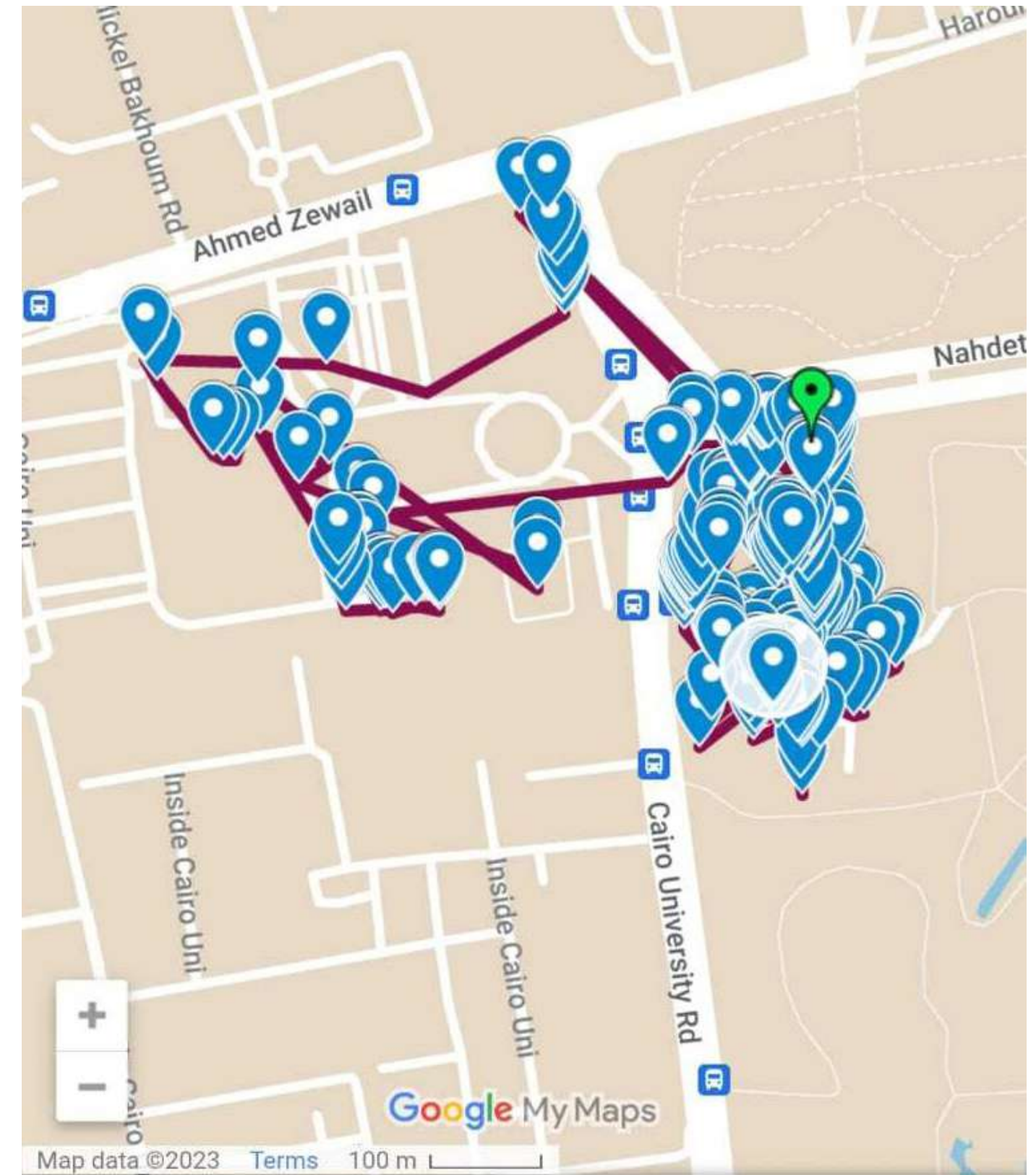
link to map: <https://www.google.com/maps/d/edit?mid=1LsOlvzaUS7RSBOBNjbBI6Xozbigj55g&usp=sharing>

link to exhibition: <https://www.forcities.org/exhibitions/for-cities-week-2023>

Following the “ForCities” exhibition, I was invited to participate in the “Death Bloom: Erased Chlorophyllic Memory” exhibition, which is part of the “SABBART” program by EUNIC Egypt.

My involvement in the “Death Bloom” exhibition allowed me to showcase my work on urban tree mapping and the digitization process. This exhibition emphasized the importance of preserving green spaces in urban environments and highlighted innovative approaches to environmental conservation.

By presenting the data and visualizations, I aimed to raise awareness about the ecological and social significance of trees in cities, and to inspire others to engage in similar initiatives.



E102



Death Bloom: Erased Chlorophyllic Memory

Art Exhibition

May 26 - June 12



The French Institute
Al Mounira

With the participation of the "Mapping
Trees in Egyptian Cities" initiative

The plants and workers of the Orman
Garden, and others

Artist: Omnia Sabry
Designer: Rana El Mahallawy
Curator: Mariam Elnozahy



Funded by
the European Union

SABBART



EU National Institutes
for Culture

REFLECTION

Two challenges arose in this project: technical and epistemological. We adopted a human-centric approach, using arm length and steps for data collection to provide an organic perspective in mapping. However, this introduced limitations and led to minor inaccuracies in location data due to mobile phone usage. Identifying and classifying tree species posed another technical challenge, which was addressed by botanical expertise. Digitizing and integrating the data into an interactive map required meticulous attention to detail and data management.

The epistemological challenge involved reevaluating our understanding and perception of trees. In the eyes of city developers, trees were often regarded as mere decorative elements or abstract representations on maps, serving as fillers in the urban planning process. However, the workshop facilitated a paradigm shift through simple acts like looking up at the trees while collecting their location data. This shift led to the realization of the immense scale difference between us and the trees. It became evident that trees symbolize not only our scale in the universe but also serve as physical evidence of their journey through time. Some of the mapped trees were more than 100 years old, representing a different era in Egypt's history when it was a powerful nation showcasing its might by introducing non-native trees and plants into the Orman Garden, creating a mini-ecosystem. Reflecting on the trees, their history, and their place in the city raised profound questions about our city's identity and our role within it.

This shift in perspective, from viewing trees as mere elements on a map or plan to recognizing them as living museums and vital components of our urban environment, was an eye-opening experience. It emphasized the importance of looking beyond surface-level aesthetics and appreciating the historical and cultural significance of trees. By looking up to the trees, we gain a deeper understanding of our city, its past, and our place within it.



02

HASSAN FATHY CULTURAL CENTRE

Final year graduation project
2018-2019

ABOUT

Graduation Project: cultural Centre in Baris oasis (Hassan Fathy's village)
the graduation project was the development of Hassan Fathy's village a project that stopped in 1967 by focusing on 4 main projects, a cultural Centre, environmental hotel, a crafts school, and an environmental research Centre. i choose the cultural Centre for many reasons the main one was me being so involved in the cultural scene in my city at that time and wanted to visualize a cultural Centre built on the principles of vernacular architecture used by Hassan Fathy and could support locals and their economies and heritage while raising awareness and connecting creatives from every field

CONTEXT

In the light of recent developments, where the historic Orman Garden, a 150-year-old green oasis in Egypt, faces closure alongside the zoo, the Mapping Trees Initiative orchestrated

□ In 1967, the visionary Egyptian architect Hassan Fathy embarked on an ambitious project in New Baris, Egypt, located at 24° 41' 36" N, 30° 35' 45" E. This project aimed to establish a self-sustaining community in the harsh desert environment, integrating urban planning, social organization, natural passive cooling, and vernacular architecture.

Hassan Fathy's Vision

Hassan Fathy (1900-1989) was an innovative architect who deviated from the neoclassical and modernist trends of his era. Drawing inspiration from ancient Egyptian architecture and vernacular design, Fathy was dedicated to addressing rural poverty by using local, affordable materials like mud bricks. His projects aimed to revitalize cultural heritage and foster community through traditional construction techniques and local materials.

Challenges and Solutions

The New Baris project began in 1963 after the discovery of a significant water source near the Kharga Oasis, which promised to irrigate about 1,000 acres of land. Tasked with building a village for 250 families, half of whom were farmers, Fathy had to create a cohesive community from scratch. He meticulously studied demographic, geographic, and climatic data, as well as ancient mud brick structures like the Bagawat Necropolis and the traditional village of Kharga. Fathy designed New Baris with a north-south axis to maximize building shadows, utilized continuous ventilation systems, and incorporated various architectural elements like courtyards, Taktaboosh sitting areas, Rasha openings, Shuksheika lanterns, and domes to facilitate natural cooling.

Community and Cooling Strategies

Fathy's holistic approach extended to community buildings, particularly the market (Suk), which became the village's social and economic hub. Equipped with wind catchers and underground storage for crops, the Suk utilized natural cooling and ventilation systems to reduce temperatures by up to 15 degrees. Water features like the Salsabil cooling plate and Nafora fountain, along with vegetation, further enhanced the natural humidification and cooling of the village.

Interrupted Ambitions

Despite the innovative design, construction was halted in 1967 due to the war between Egypt and Israel and new regulations on earthen buildings. Only a few structures were completed, including the market, district administration building, outer shell of the museum and social center, and several villas. Nonetheless, Fathy's work in New Baris remains a testament to his ingenuity in creating naturally ventilated, community-focused architecture in challenging environments.

My Graduation Project

Building upon Fathy's legacy, my graduation project involved designing a cultural center in Baris Oasis. This project aims to continue Fathy's vision, blending traditional techniques with modern needs to create a sustainable and culturally significant community space.



Site Visit

IDEATION

Concept Development

After the site visit, I began conceptualizing the cultural center by reflecting on the environment and the community's needs. I decided not to mimic Fathy's architectural style directly, but rather to implement the core principles he used—such as natural ventilation, sustainable materials, and cultural sensitivity—in a different style. This led me to use large rocks as building elements and tents for shading, inspired by traditional Bedouin shelters. These elements not only offer protection from the intense sun but also blend seamlessly with the natural landscape.

Community and Cultural Integration

Understanding the social dynamics of the area, I integrated spaces that foster community interaction and cultural activities. The center features exhibition halls, workshop areas, and open spaces for performances and gatherings. These elements are designed to encourage community engagement and cultural exchange, reinforcing the identity and heritage of the region.



Site Visit



Site Visit
22



Site Visit

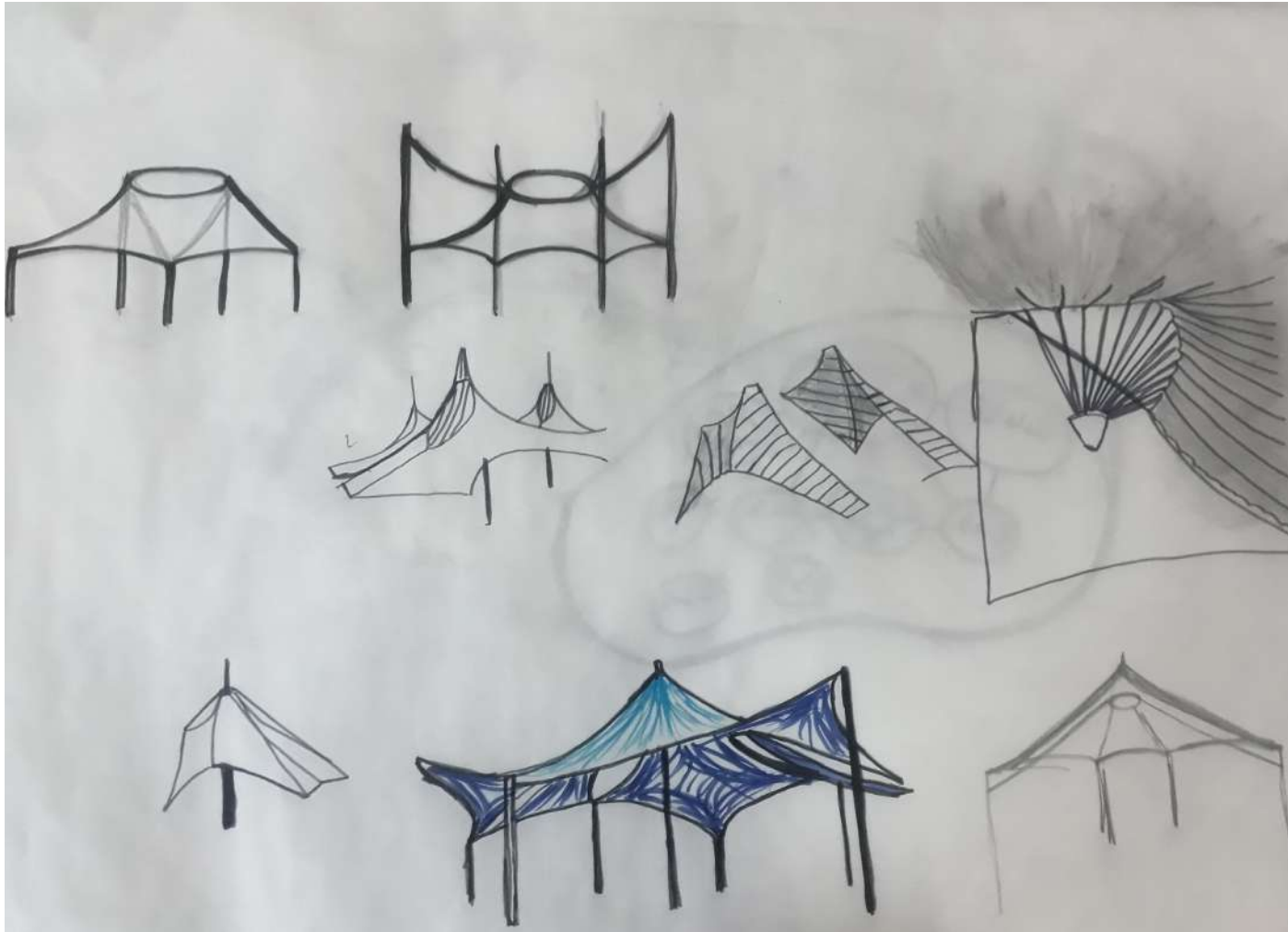
IDEATION

Concept Development

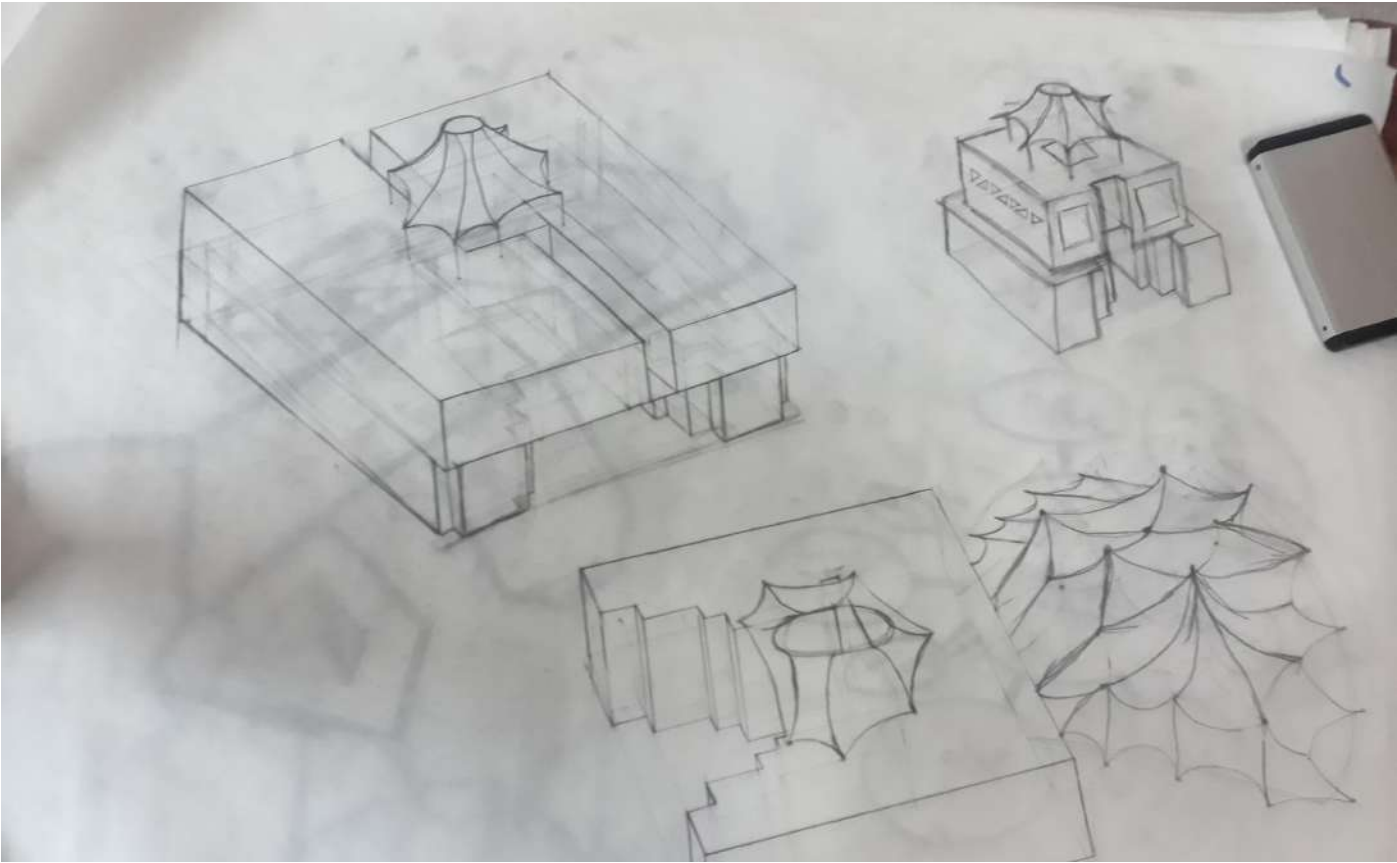
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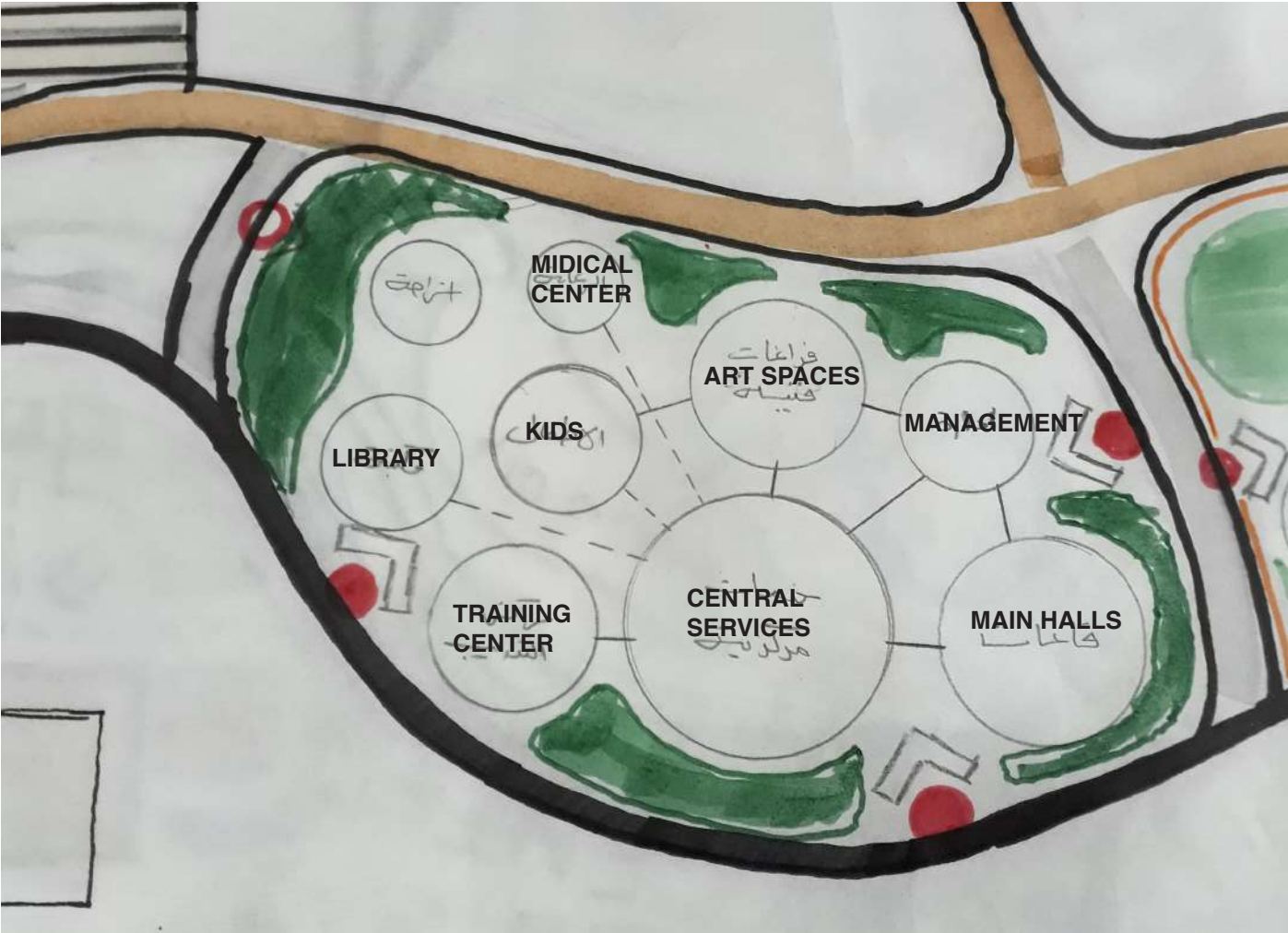
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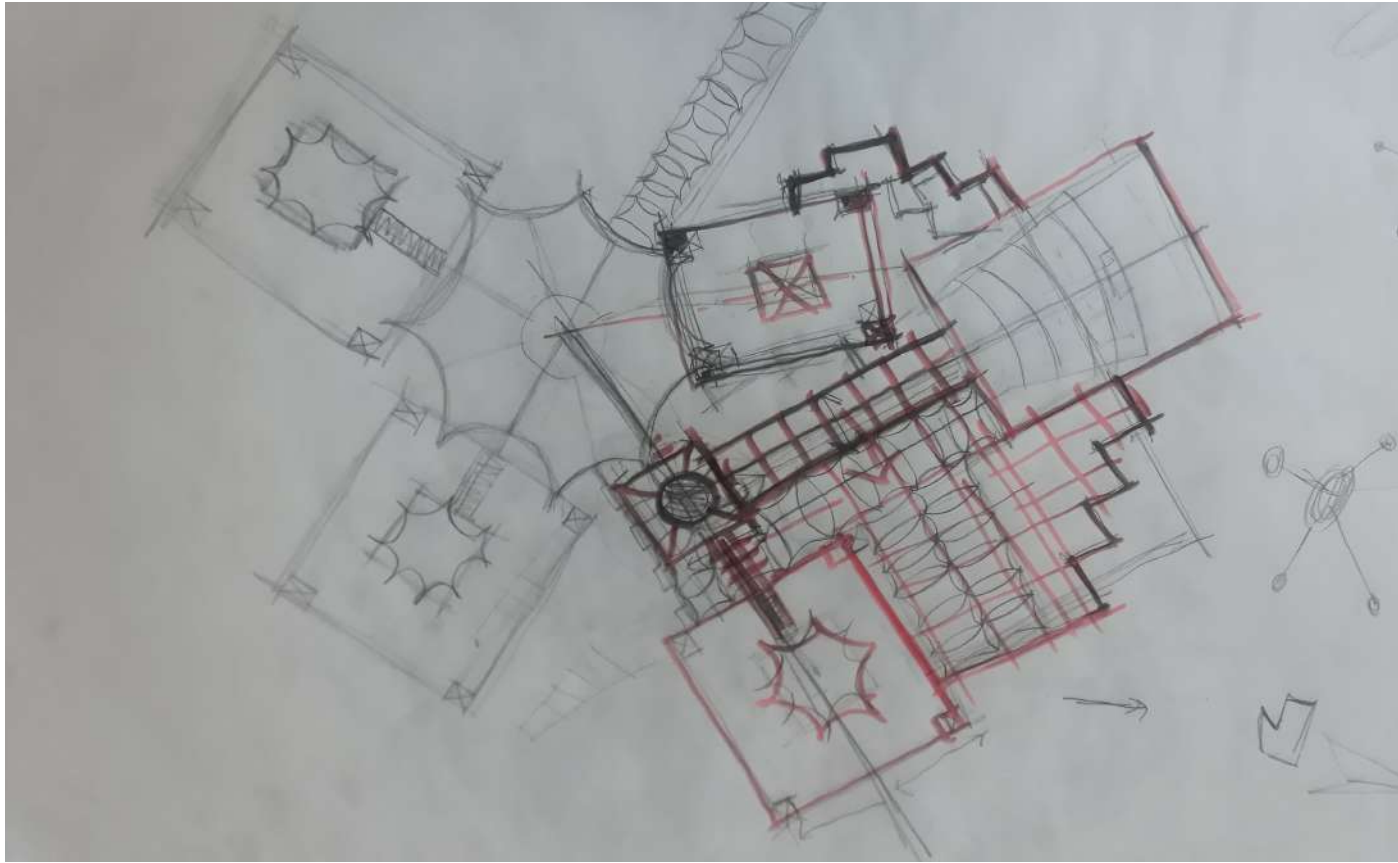
Concept Development
24



Concept Development



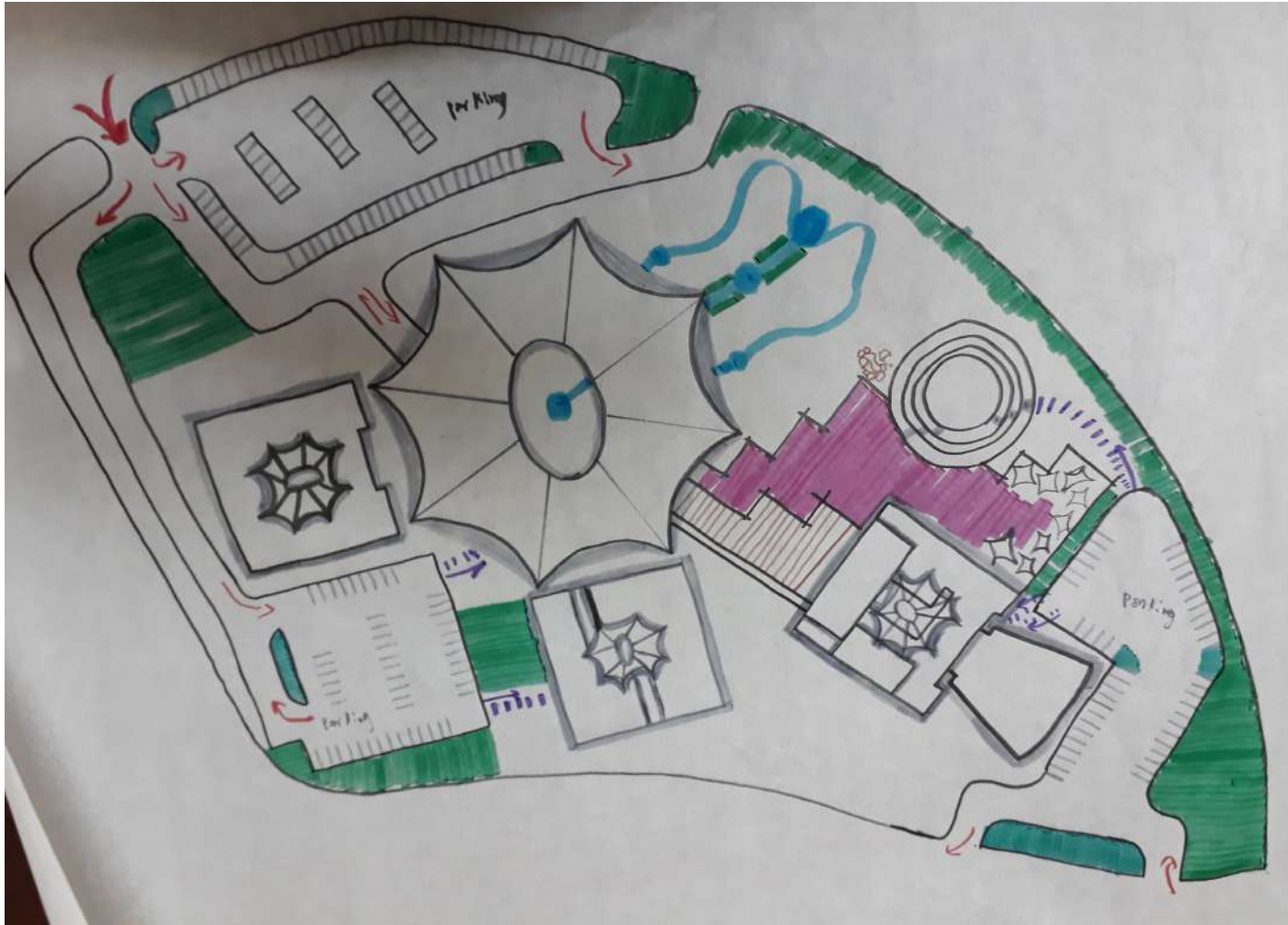
Concept Development



Concept Development



Concept Development



Concept Development
26



Concept Development

IMPLEMENTATION

Design Approach

My design for the cultural center incorporates innovative architectural elements tailored to the desert environment. The structure is shaped like a reverse pyramid, which effectively limits heat absorption and enhances natural ventilation. To provide shade, I used large rocks and tents, inspired by traditional Bedouin shelters. These elements not only offer protection from the intense sun but also blend seamlessly with the natural landscape.

Natural Ventilation and Cooling

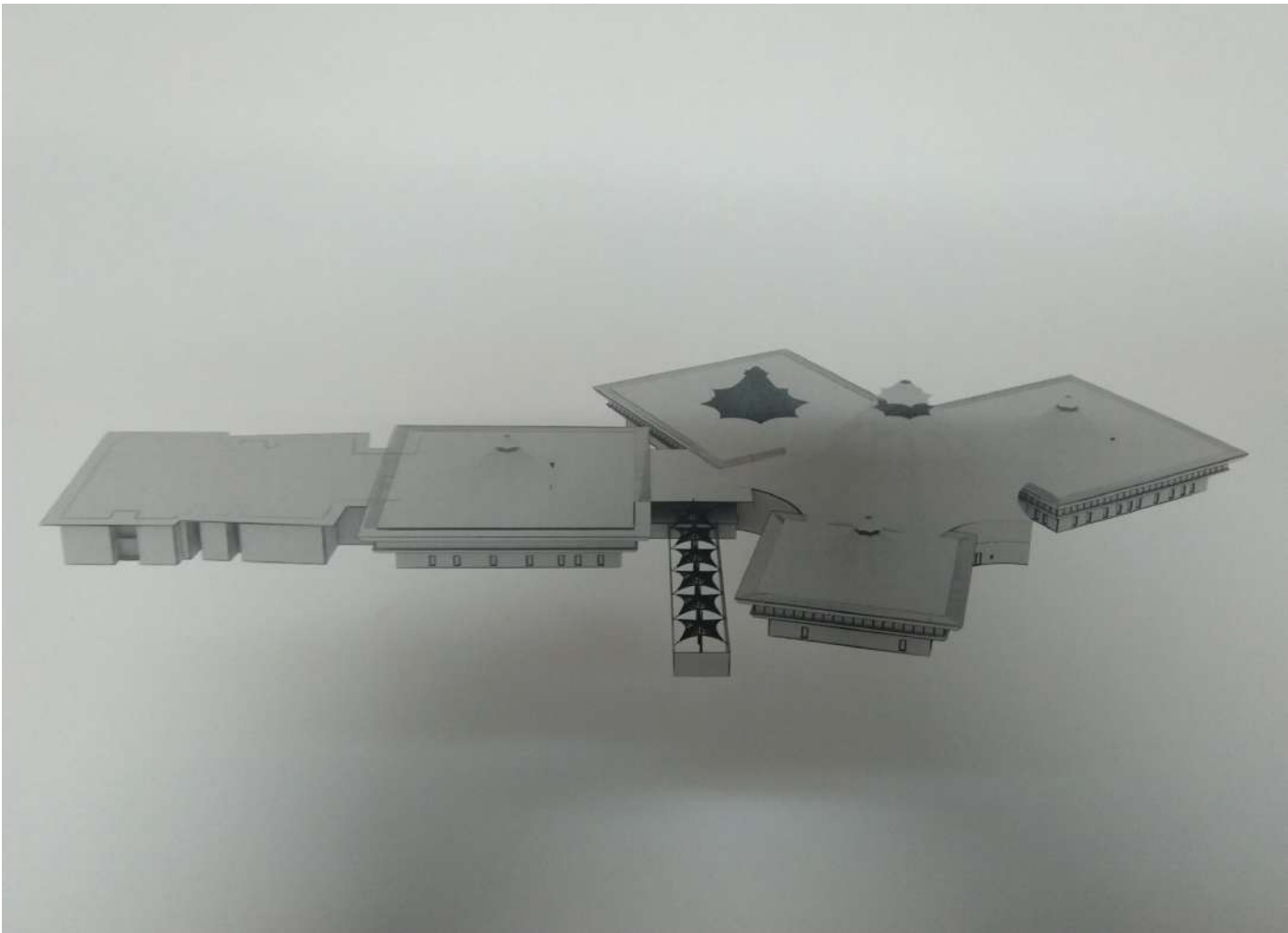
A key feature of my design is the use of wind catchers, strategically placed to harness and direct the wind through the building. This natural ventilation system ensures a continuous flow of air, keeping the interiors cool without relying on artificial cooling methods. The reverse pyramid shape further aids in maximizing shade and promoting airflow, making the structure both energy-efficient and comfortable.



Concept Development

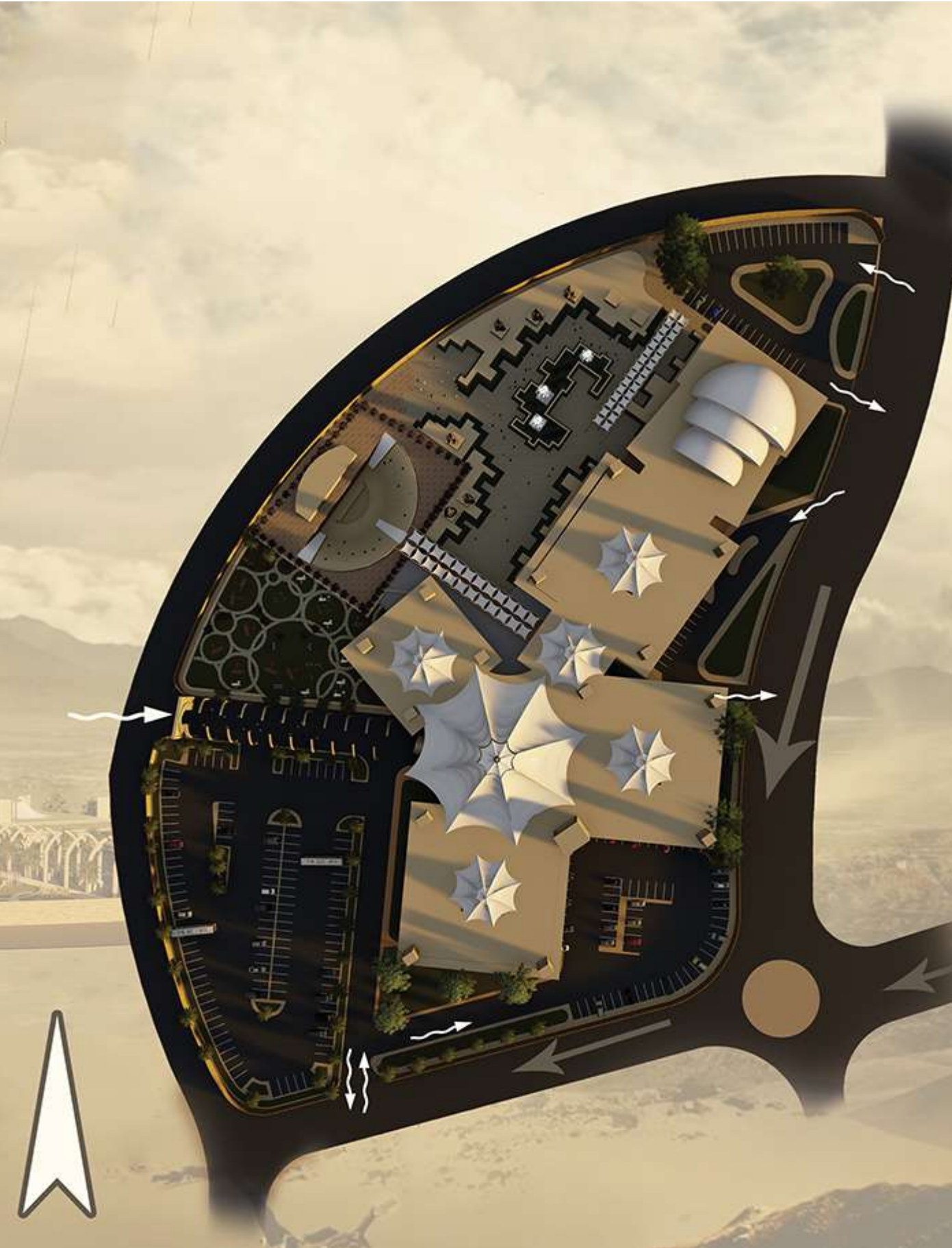


Concept Development
28



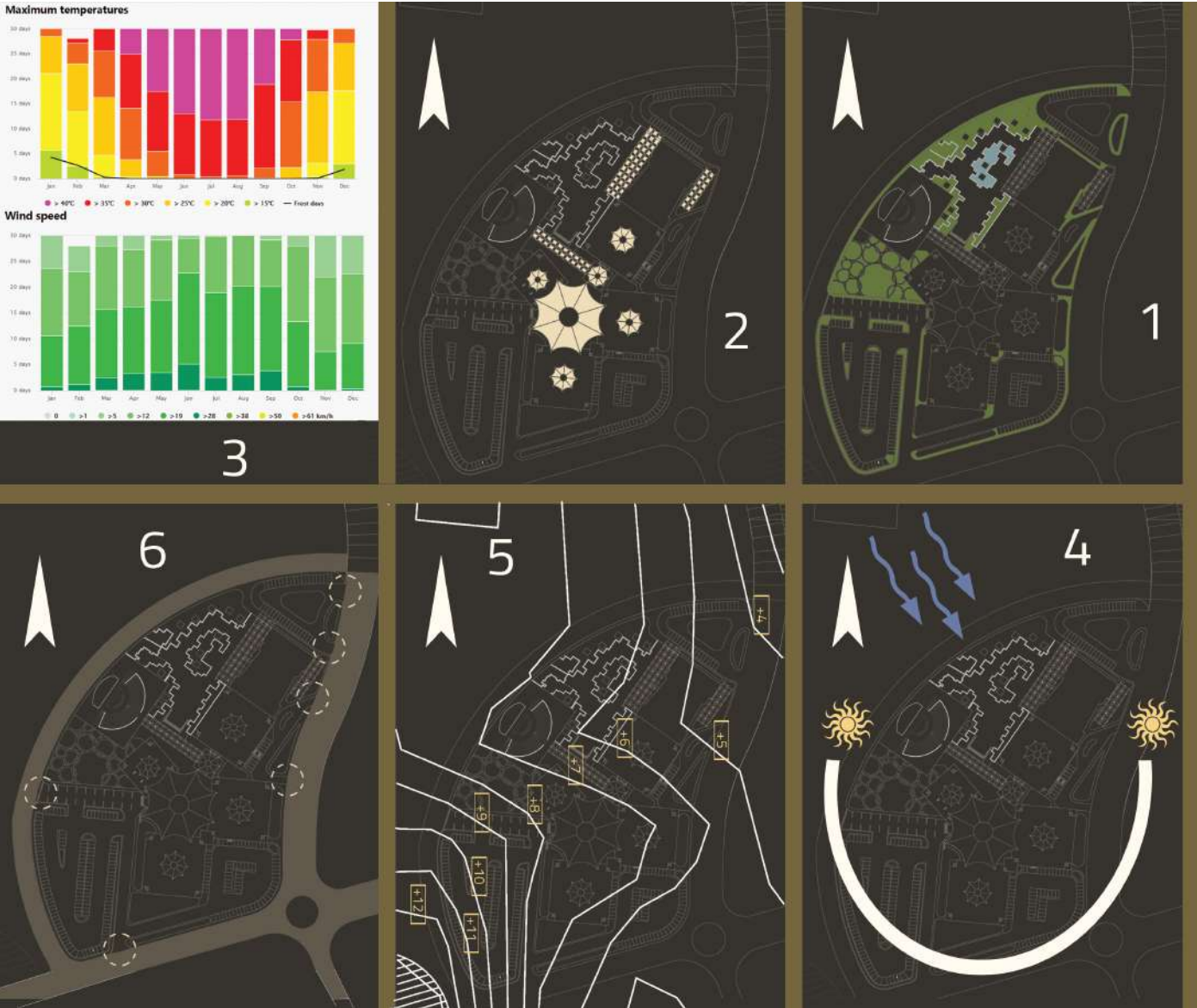
Concept Development

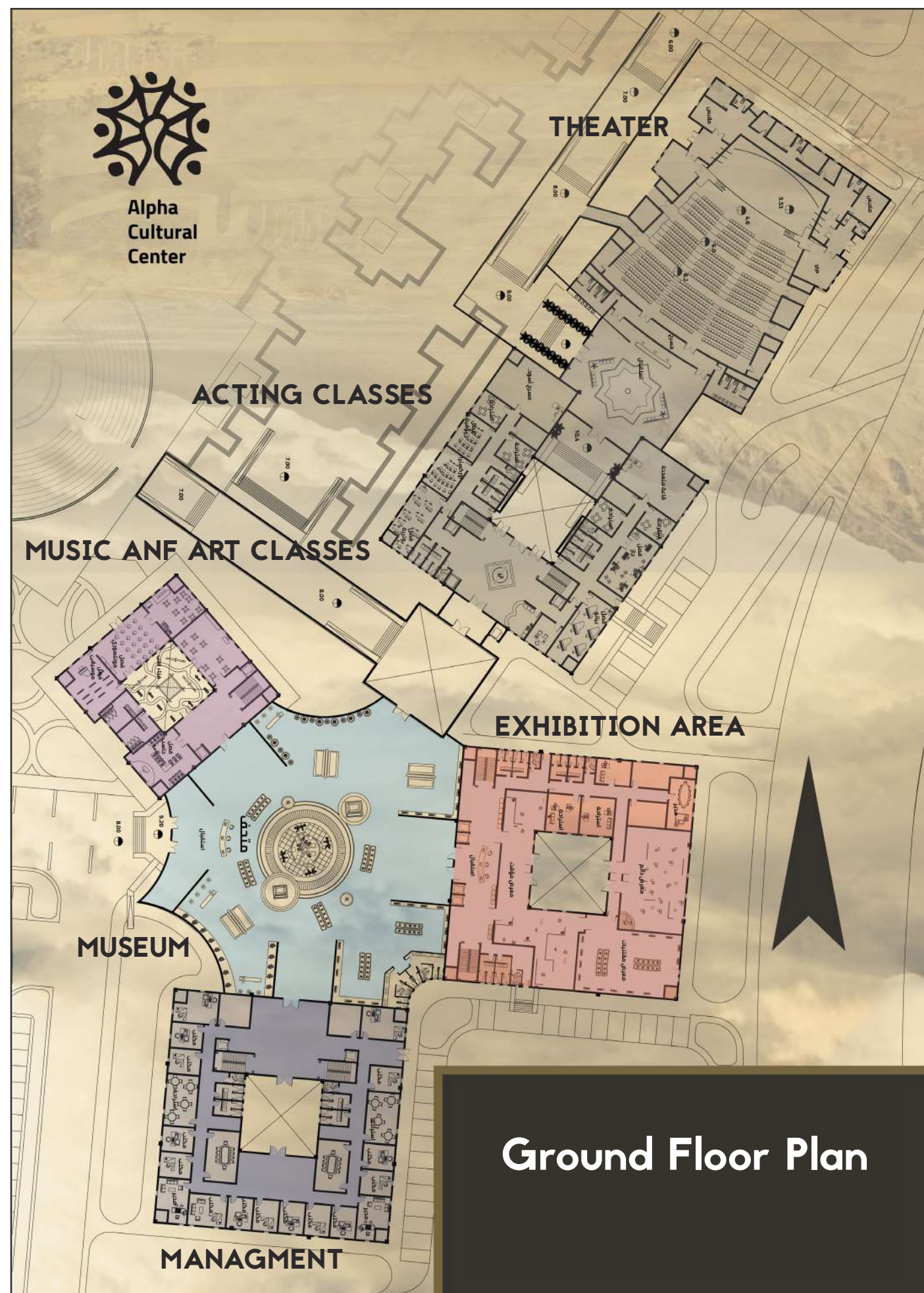
Layout



ANALYSIS

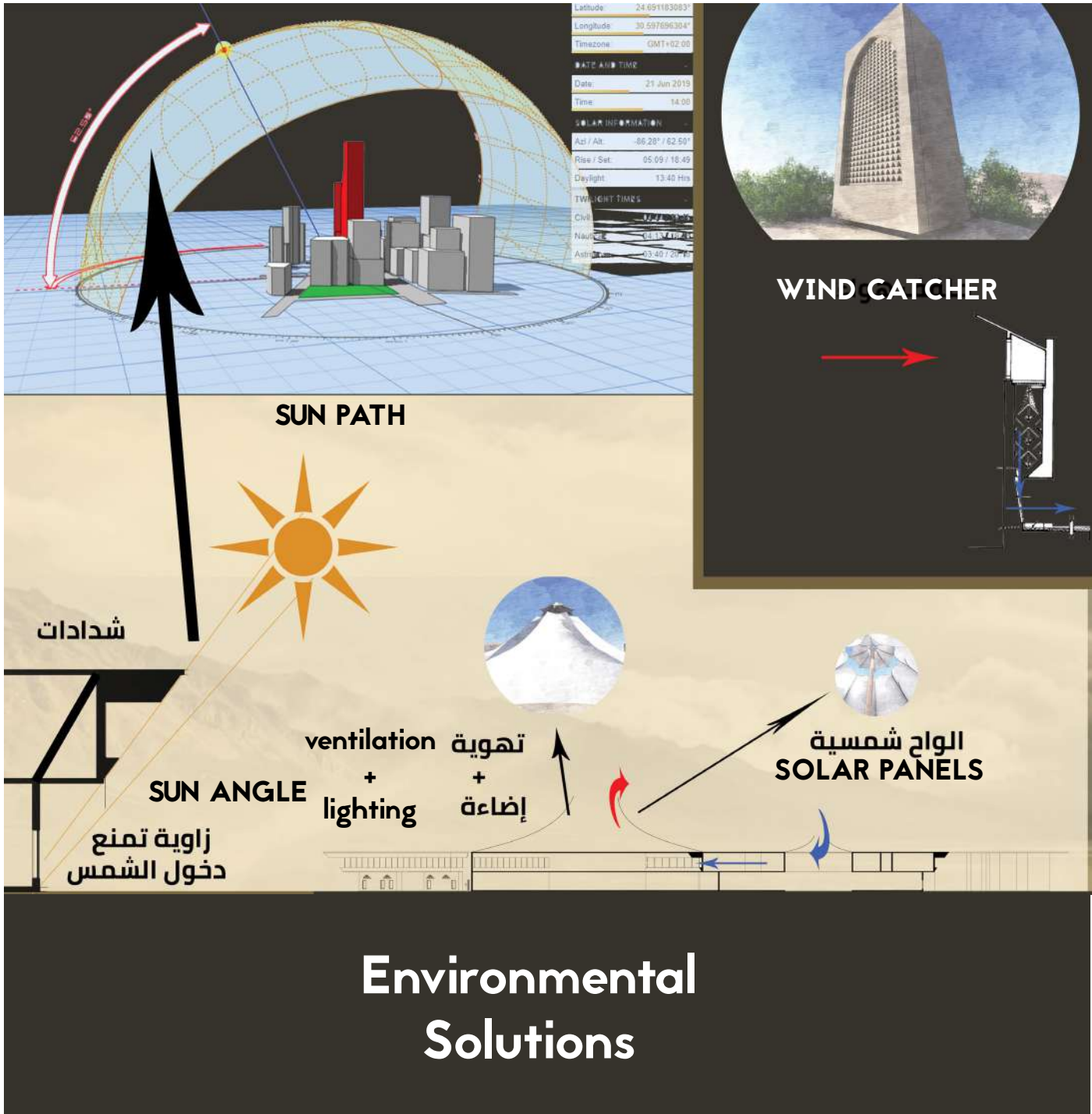
- 1- GREEN ELEMENTS
- 2- SHADING ELEMENTS
- 3- TEMPRATURE AND WIND SPEED
- 4- SUN PATH AND WIND DIRECTION
- 5- CONTOUR
- 6- ROADS AND ENTRANCES





LOGO DESIGN CONCEPT

Inspiration



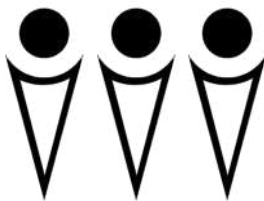
Intangible Cultural Heritage



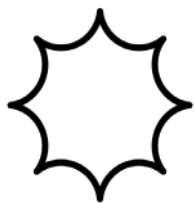
United Nations Educational, Scientific and Cultural Organization



World Heritage Convention



People



Tent



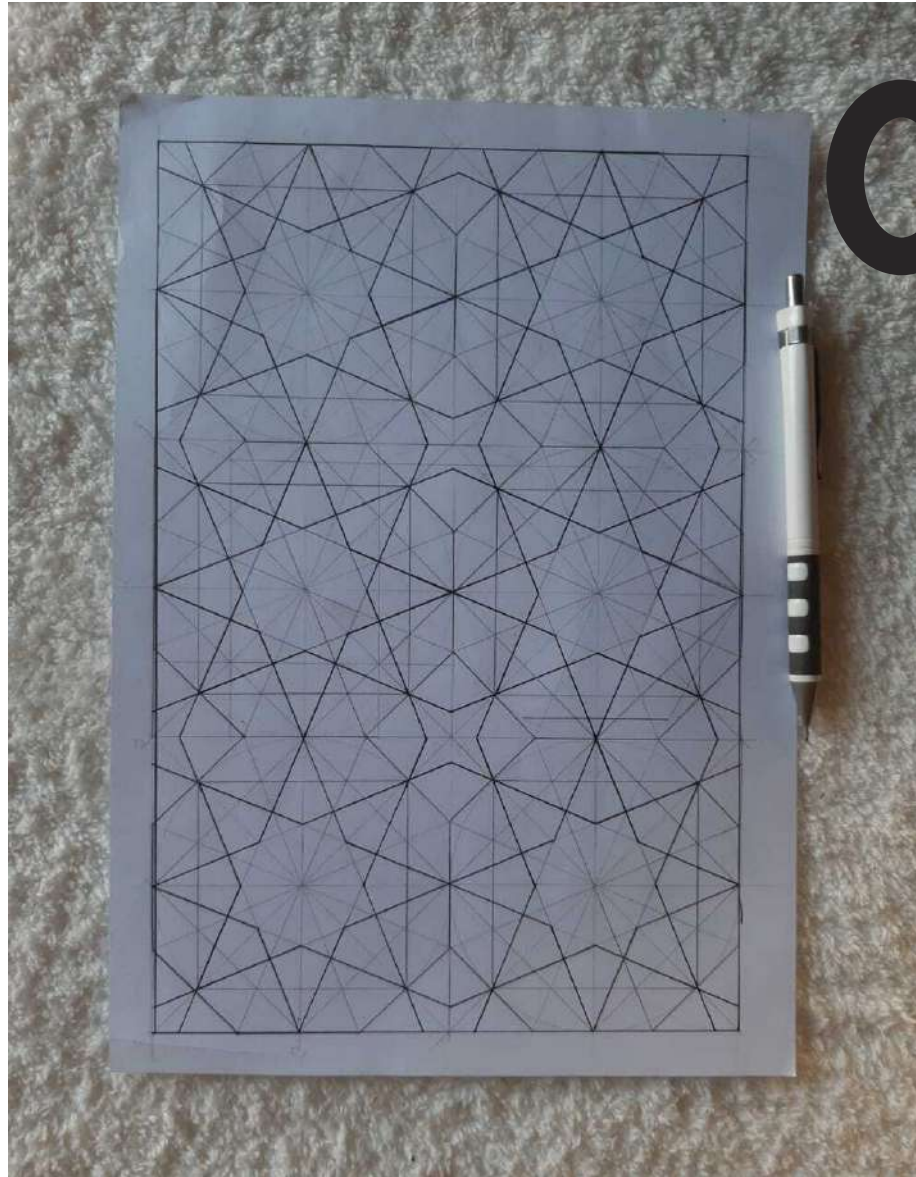
logo

REFLECTION

For me this graduation project was more than an academic exercise; it was a journey of exploration and learning. It reinforced my belief in architecture's power to harmonize with nature, celebrate diversity, and enrich communities. Moving forward, the lessons learned and the experiences gained from this project will undoubtedly shape my approach to future architectural endeavors, with a steadfast commitment to sustainability, cultural sensitivity, and human-centric design.

Designing a cultural center in Baris Oasis has been a profoundly transformative experience, influencing both my professional growth and personal development. This project enabled me to explore sustainable architecture principles and community-centric design in depth, significantly impacting my career choices.





03

IDEATION

For the design, I conceived the idea of using a family of Islamic geometries that utilize rectangular shapes, aiming for ease of use and assembly while maintaining artistic appeal, especially for children. In Cairo, a city renowned for its artistic ornaments, I embarked on tours of historical Islamic buildings. My intention was to use these patterns to recreate the famous Ramadan lantern for the children in Ard El-Liwa, engaging them in the process. I meticulously traced patterns during these tours, which served as inspirations and aids for the project.

FABRICATIONS

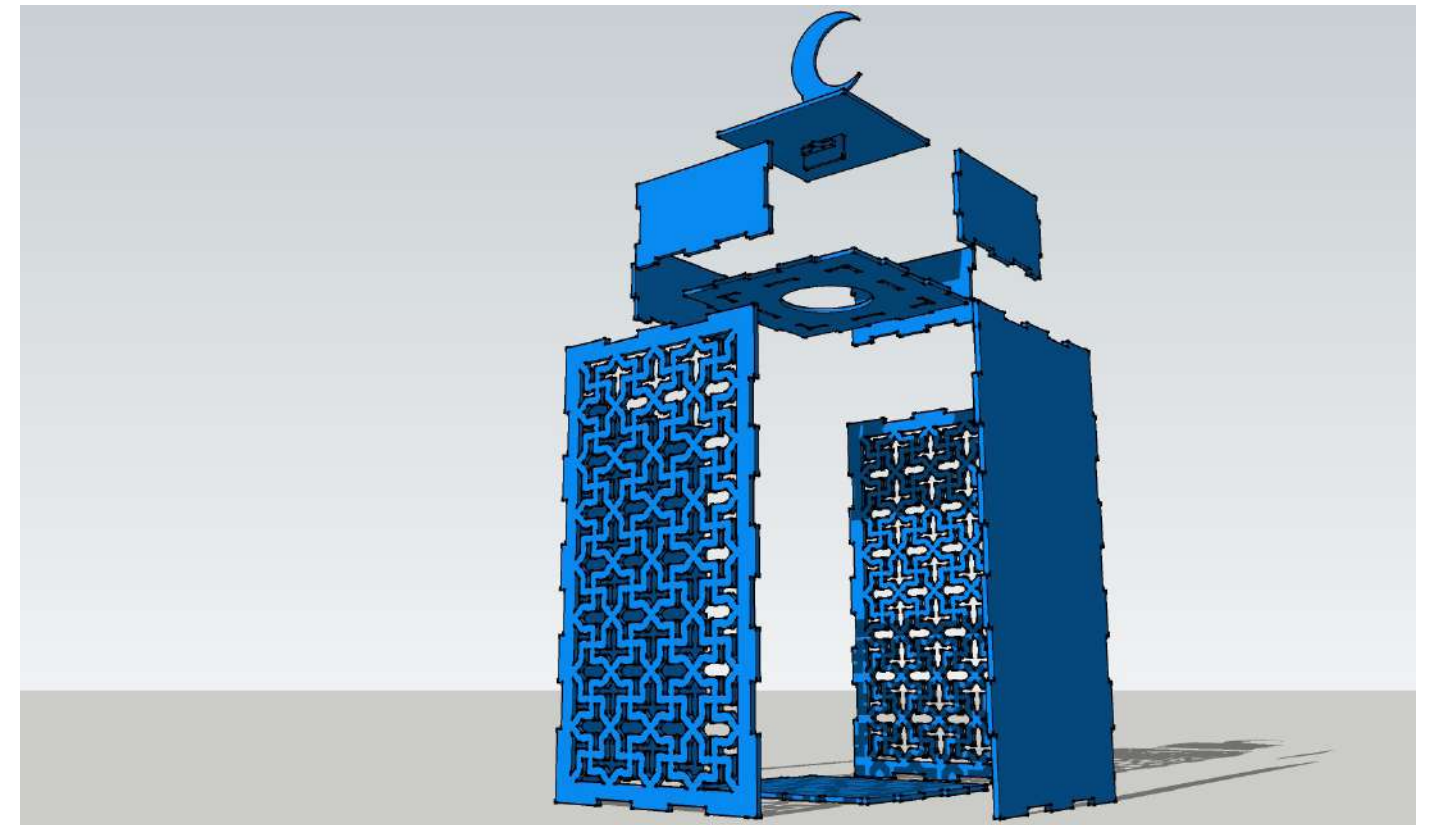
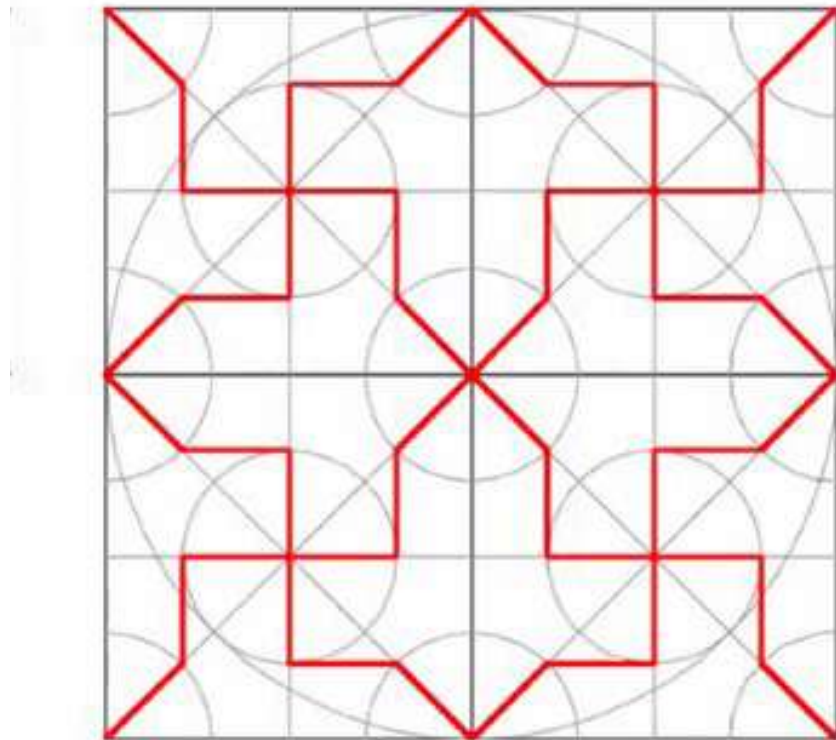
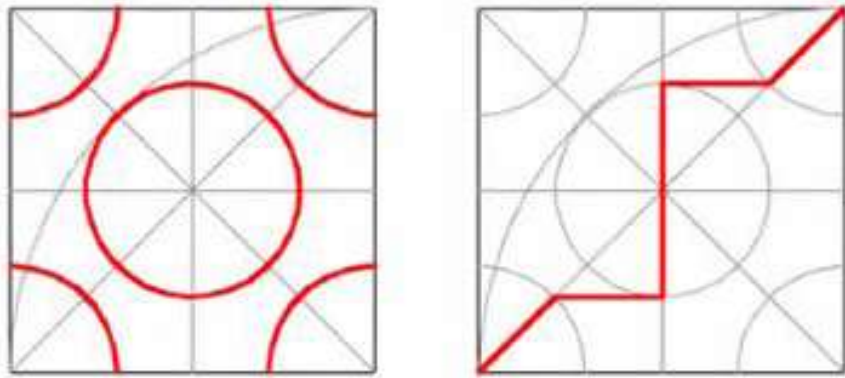
CONTEXT

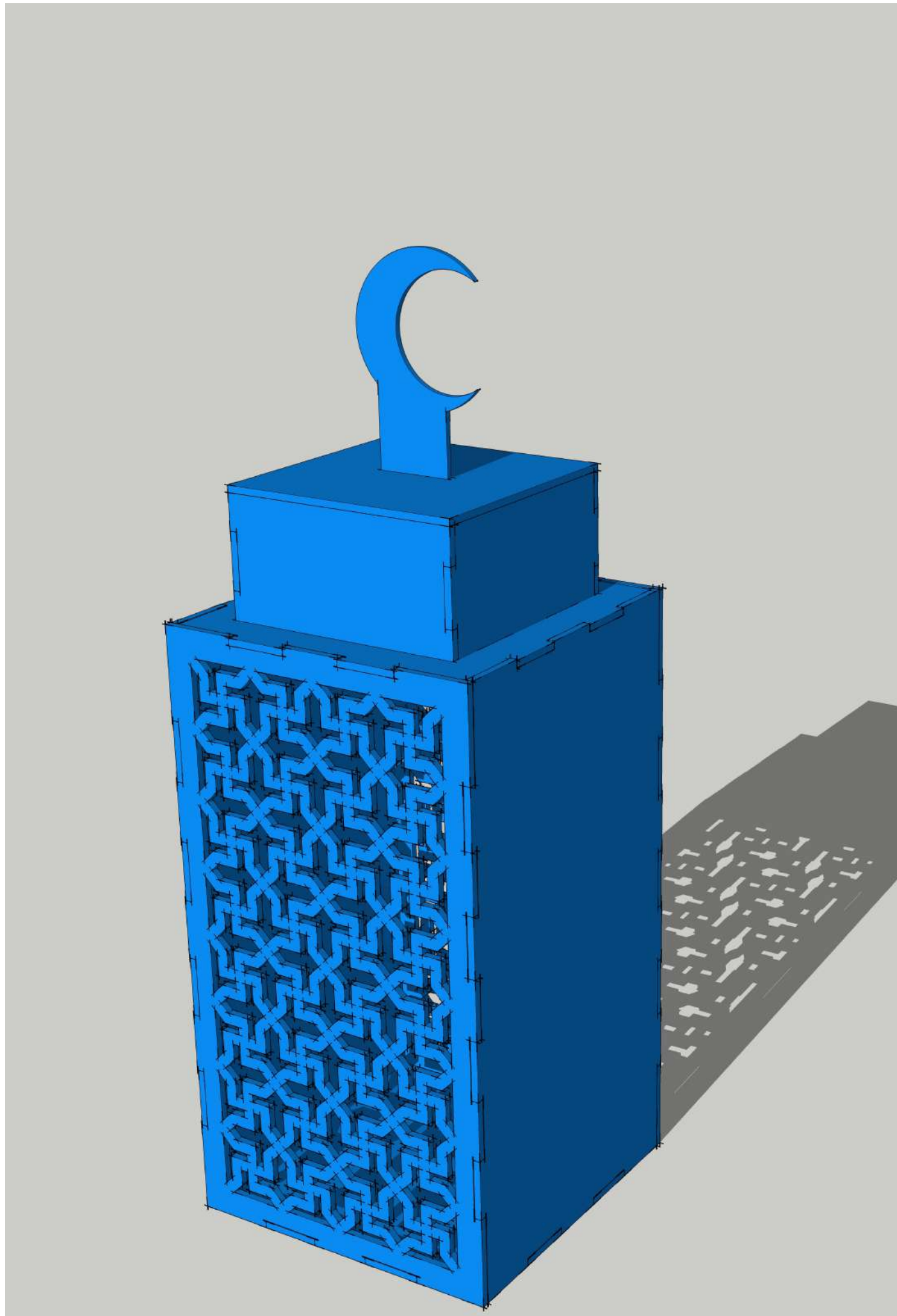
As the Fabrication Lab Coordinator at CLUSTER, I was responsible for organizing community workshops within the informal neighborhood of Ard El-Liwa. During one workshop held in the month of Ramadan, I took the initiative to redesign the traditional Ramadan lantern, a significant symbol in the month's rituals.



IMPLEMENTATION

After identifying and tracing the patterns, I envisioned the final product and prepared detailed drawings and materials needed for the design. Utilizing a laser cutting machine, I precisely cut the designs. Finally, I worked with the children to assemble the lanterns, fostering their participation and creativity in bringing the project to life.









MAPPING ASYUT URBAN HERITAGE

ABOUT

I was honored to collaborate with The National Museum of Egyptian Civilization as an architect representing Asyut, Upper Egypt. My contribution involved mapping the urban heritage of Asyut and creating visualizations for an exhibition highlighting the city's profound historical significance. This project allowed me to showcase my expertise in architectural representation and heritage visualization, underscoring my commitment to preserving and celebrating cultural legacies through design.

CONTEXT

Asyut, situated in Upper Egypt, holds a deeply significant place in my life—it's not only my birthplace but also a city I've come to cherish profoundly over the years. My attempt to raise awareness about its rich history and document its urban heritage is a personal mission. I had the privilege to exhibit this ongoing project at The National Museum of Egyptian Civilization, showcasing Asyut's cosmopolitan history alongside its cultural and architectural significance.

To begin, I conducted comprehensive site visits to categorize and select key sites of historical and cultural importance. Asyut's cosmopolitan heritage is marked by its role as a cultural crossroads, influenced by various civilizations including ancient Egyptian, Greek, Roman, Coptic, Islamic, and more recent influences. In the 19th and 20th centuries, Asyut thrived as a center of trade, education, and cultural exchange, attracting diverse communities and fostering a rich tapestry of traditions and architecture.

This diversity is evident in Asyut's architecture, which blends traditional Egyptian styles with elements borrowed from other cultures, creating a unique cosmopolitan aesthetic. The city's vibrant marketplaces, educational institutions, and religious sites reflect its dynamic history and ongoing cultural vitality.

Exhibiting this project at the museum marks a significant milestone in my journey to document and celebrate Asyut's heritage. For me, it's not merely a concluded effort but an ongoing commitment to preserve and promote the city's profound legacy.



19th century water pumping station



Alexan Pasha Palace



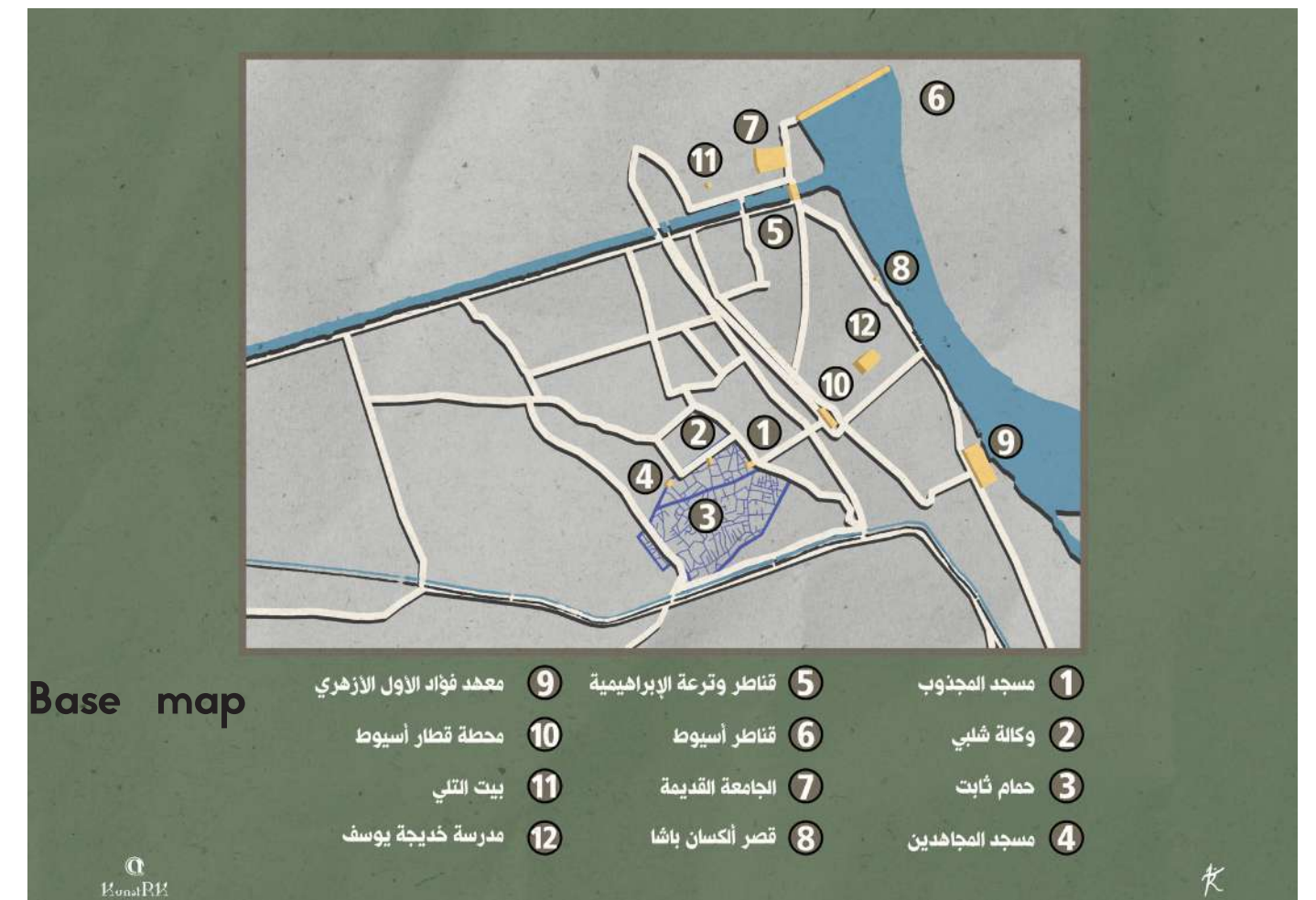
The Old College

IDEATION

After selecting the locations, I started researching old maps of the city and delving into literature about the chosen sites to enhance my understanding and presentation.

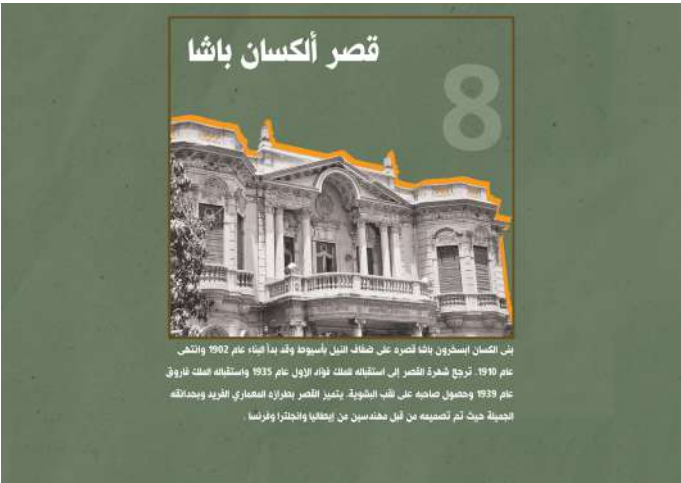
I located a map from 1826, meticulously tracing and geolocating it to align with modern maps. This process allowed me to track the city's urban evolution and accurately map out the selected places using GIS.





IMPLEMENTATION

After completing the mapping phase, I began selecting how to present my work within the constraints of the museum and started creating graphic design panels for printing and exhibition.

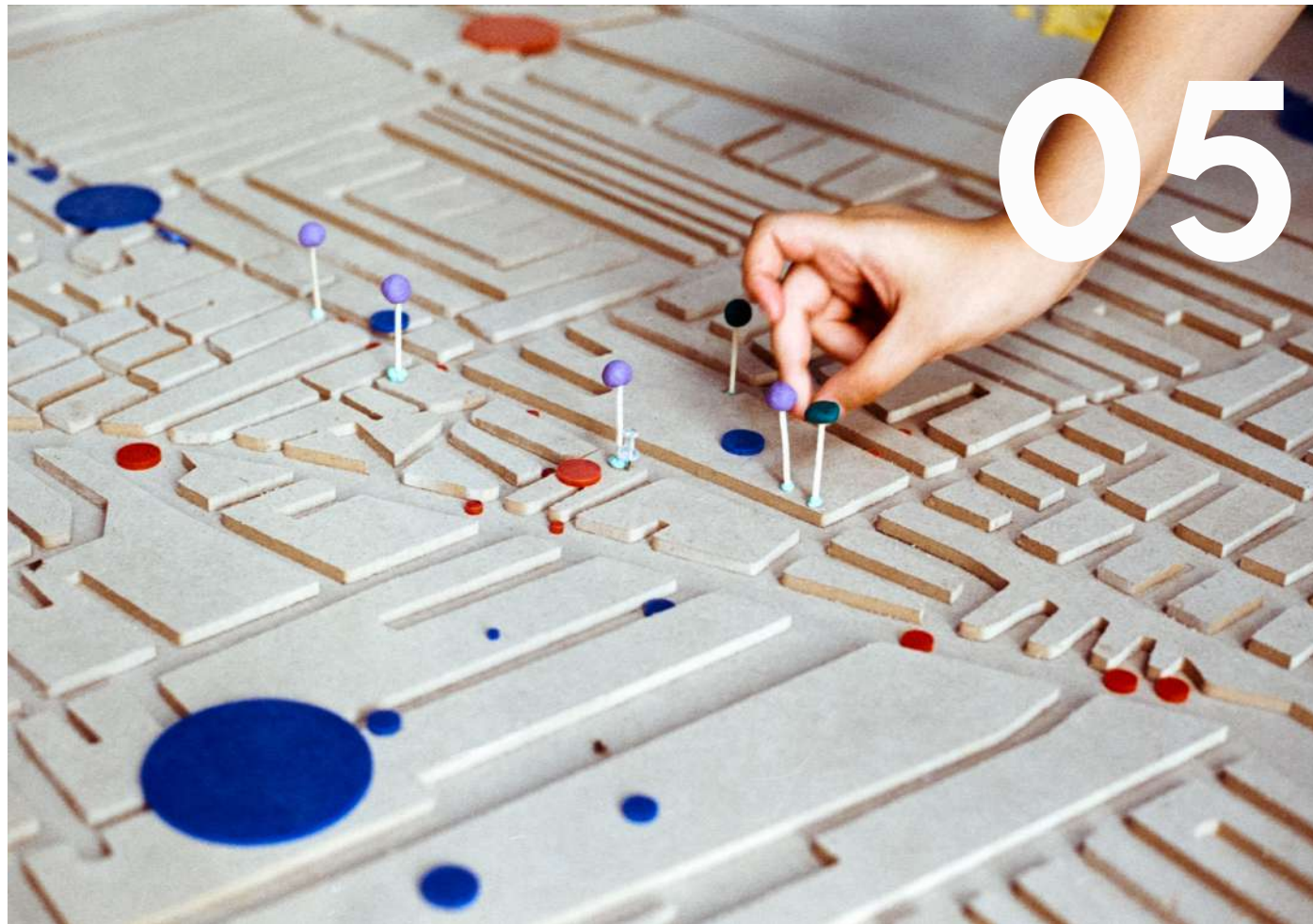


Qanater Ibrahimia, is a series of water bridges and dams located in Egypt. It was built in the 1860s to regulate the flow of water from the Nile River into the irrigation canals of the Nile Delta. The construction of Qanater Ibrahimia played a crucial role in the development of Egypt's agriculture and economy.

قناطر وترعة الإبراهيمية



تعد ترعة الإبراهيمية أحد أعظم المنجزات الهندسية في مجال الري في مصر الحديثة حيث بدأ العمل عليها من عام 1867 وانتهى العمل بها عام 1873 وتمتد الترعة من محافظة أسيوط إلى محافظة بني سويف حيث يبلغ طولها 267 كم واشتغل في حفرها نحو مائة ألف من العمال المصريين يدوياً . وقد كان هذا المنجز عملاً مصرياً خالصاً حيث تم تصميم وتنفيذ الترعة عن طريق عمال ومهندسين مصريين.



CONTEXT

The research in Cairo was inspired by the concept of sensory urbanism, which encourages thinking and designing cities through sensory information. Given that contemporary urban design heavily depends on function and visual aesthetics, reviving an appreciation for neglected senses, such as sound, could provide important insights into the past and future of cities. After several discussions, soundscape and sound mapping were chosen as suitable methods to capture the vibrance and everyday life of the megalopolis.

Cities are filled with various sounds and rhythms, from railroad crossings to the swaying of leaves and the rhythm of people passing by. These sounds and rhythms appear and disappear based on space and time. The sound and musical environment surrounding us, combined with visual information, physical sensations, and our mental states, act as a metronome, guiding us through the cycles and patterns of a city.

SOUNDMAPPING

ABOUT

The collaboration between ForCities and CLUSTER began with a series of conversations about the future of cities and how they could be analyzed beyond cultural boundaries. ForCities, coming from Japan where urban depopulation and economic stagnation have become the norm, was fascinated by the rapid growth of Cairo. This led to a keen interest in understanding how urban actors, especially those working from the bottom-up, are transforming the urban landscape of this dynamic megalopolis.

In July 2022, this collaboration culminated in a two-day pop-up exhibition and nine workshops in Cairo, Egypt. The collaborative efforts involved various participants, including architecture students, designers, artists, urban explorers, citizens, and children from one of Cairo's informal areas. The collaboration also featured Elizaveta Belyakova from GIS FRIENDS, who joined from Moscow.



IDEATION

The concept was to create a collective experience where participants could engage with the city's sounds, visualizing and interpreting them in innovative ways. The project was divided into several parts, each designed to explore different aspects of urban soundscapes. The idea was to not only present these sounds but also to engage the community in creating and interacting with them. This included sound mapping, sound donations, and artistic compositions inspired by the neighborhood's unique auditory landscape.



IMPLEMENTATION

As part of the CLUSTER team, I was hands-on in bringing these ideas to life:

1- Soundmapping:

We created a CNC machine-cut map of the Ard al-Liwa neighborhood in Alfabrika. This map served as a collective canvas where visitors could add layers representing their interpretations of the neighborhood's sonic aspects.



2- Urbanist Kit / Audible Cities:

We designed and developed an educational tool called the Urbanist Kit, which included various mini-exercises to help participants explore and understand the sounds of the city. This kit encouraged co-imagining what our cities could be through alternative soundscapes.



3- Sound Observations:

The exhibition featured a series of posters, photo journals, and research findings from the CLUSTER team. My contribution, titled "If Cities Could Speak," reimagined the sounds of Salah al-Din Square during Cairo's peak historical period, aiming to preserve a part of the city's auditory memory.



"If Cities Could Speak"

By Khaled Ragab

In the past few years, Cairo has witnessed drastic changes in its image, and throughout the process of expanding, building and demolishing, the city is being redefined and reshaped. In that process of constant change there is always the fear of losing the sense of the city and how it really felt to live in the great city called Cairo.

So, this is my attempt to imagine how the city sounded at the peak of its glory. I chose the spot of Salah al-Din Square, which was known as Alrmylh Square back when Cairo was one of the most important cities in the world, when that square was the center of the city. Back then, it wasn't the same touristic, almost-dead area with the sound of cars more obvious than any other sound, it was actually the complete opposite: it contained a huge market for cattle and horses, it hosted military parades and visiting regiments, and it was surrounded by the school and mosque of Al-Sultan Hassan — one of the greatest buildings in Islamic history — and Salah al-Din's castle built on the mountain nearby. So, I will try to hear how this combination of people, animals and nature might have sounded and hopefully it can save a part of the city's memory.

QR: <https://drive.google.com/drive/u/7/folders/1W3nk1NKtpdgtSgnDO7laL3jRGnc7BsO9>





The Square Currently

الميدان حالياً



The Square In The Past

الميدان قديماً



The Imagined Square

الميدان المتخيل



REFLECTION

This exhibition was a significant milestone in my career, developing my ability to blend urbanism with sensory experiences to create a deeper understanding of city life. It also made me focus on the role of non-human actors in the city and opened the door for future collaborations with ForCities, such as the Green Atlas interactive map.





The Field Trip was to Chatby Tombs which is a very interesting site from the greek-roman era because it had tombs for people from a very diverse social classes which was not common at these times.
The site was unfortunately very neglected and in a poor state which is sad but it was perfect for a case study of how to manage such a very rich and challenging site at the same time.

PERSONAL PROJECTS

ABOUT

HERITAGE MANAGEMENT PRINCIPLES: By "Heritage For All"

Intensive Training Course: Heritage Management Principles
"HeritageForAll" initiative has the pleasure to inform you that for the first time, we will organize under the title "Heritage Management Principles". It defines the features of managing the heritage sites and the world heritage sites through these thematic discussions: Heritage Management Process, Legislation of Heritage Conservation, and Fundraising and Financing.





ABOUT

Visual Philosophy:
By "Isaona Kherrou"

Visual Philosophy is an experimental concept that blends philosophical reflection with artistic production. This innovative class invites participants to engage in collective discussions about basic philosophical notions. Together, we explore the foundations of each concept, discuss their applications to our individual and collective realities, and transform our understanding into visual art through colors, shapes, and textures. The goal is to build a bridge from your unique thought processes to your unique way of visualizing the logic you construct while thinking. No technical skills are required, just a desire to explore your inner world, get inspired, and inspire others.

this course helped me visualize abstract concepts into shapes and forms which was one of main goals attending the course.

